

Transmission Spectra in EXHALE_transit.py:

$H\alpha/H\beta$ from the Non-LTE $n = 2$ Population,

and the Metal Resonance Doublets

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1. Overview

EXHALE_transit.py (the Transmission Probability Model shipped with EXHALE) computes transit transmission spectra by integrating the line-of-sight (LOS) optical depth on an impact-parameter grid, using Voigt profiles evaluated through the Faddeeva function, and then averaging over the projected stellar disk with optional instrument and planet-rotation convolutions. The original code handles the He I 2^3S triplet (10830 Å) and the H I / D $Ly\alpha$ doublet (1215.67 Å).

This memo documents the addition of the $H\alpha$ (6562.8 Å, $n = 2 \rightarrow n = 3$) transmission spectrum. Unlike $Ly\alpha$, which absorbs out of the ground state ($n = 1$) that traces the bulk of the atomic hydrogen, $H\alpha$ absorbs out of the $n = 2$ level, whose population must be computed from a non-LTE balance. We follow **Christie, Arras & Li (2013, ApJ 772, 144)**, including collisional excitation/de-excitation, recombination cascade, two-photon decay, ℓ -mixing, and **$Ly\alpha$ radiative pumping** ($1s \leftrightarrow 2p$). The $Ly\alpha$ mean intensity $J_{Ly\alpha}(r)$ is *not* produced by EXHALE; it is either read from an external text file or, by default, estimated with the simple prescription of Huang et al. (2017) — see §2.2.

The H α branch always runs; only the source of $J_{\text{Ly}\alpha}$ depends on whether `Jlya_file` is set. The existing He/Ly α behavior is unchanged.

This memo also documents the **metal resonance doublets** (Mg II h&k, Ca II H&K, Na I D; §6.), which are computed by the same LOS pipeline and produce the same three-curve spectrum figures as the H/He lines.

2. The $n = 2$ hydrogen population (Christie et al. 2013)

The $2s$ and $2p$ states are treated separately because they have very different lifetimes (the $2p \rightarrow 1s$ Ly α decay is fast, $A_{2p \rightarrow 1s} = 6.3 \times 10^8 \text{ s}^{-1}$, whereas the $2s \rightarrow 1s$ two-photon decay is slow, $A_{2s \rightarrow 1s} = 8.26 \text{ s}^{-1}$). With $n_{1s} = n_{\text{HI}}$, n_e and T taken from the EXHALE profiles, the rate-equilibrium equations are (Christie et al. Eqs. 12–13)

$$[A_{2p \rightarrow 1s} + B_{2p \rightarrow 1s} J_{\text{Ly}\alpha} + (C_{2p \rightarrow 1s} + C_{2p \rightarrow 2s})n_e + \Gamma_{2p}] n_{2p} - C_{2s \rightarrow 2p} n_e n_{2s} = (B_{1s \rightarrow 2p} J_{\text{Ly}\alpha} + C_{1s \rightarrow 2p} n_e) n_{1s} + \alpha_{2p} n_e^2, \quad (1)$$

$$[(C_{2s \rightarrow 1s} + C_{2s \rightarrow 2p})n_e + \Gamma_{2s} + A_{2s \rightarrow 1s}] n_{2s} - C_{2p \rightarrow 2s} n_e n_{2p} = C_{1s \rightarrow 2s} n_e n_{1s} + \alpha_{2s} n_e^2. \quad (2)$$

This is a 2×2 linear system in (n_{2p}, n_{2s}) ,

$$\begin{pmatrix} L_{2p} & -C_{2s \rightarrow 2p} n_e \\ -C_{2p \rightarrow 2s} n_e & L_{2s} \end{pmatrix} \begin{pmatrix} n_{2p} \\ n_{2s} \end{pmatrix} = \begin{pmatrix} S_{2p} \\ S_{2s} \end{pmatrix},$$

solved per radial cell by Cramer's rule. The total $n = 2$ density driving H α absorption is $n_2 = n_{2s} + n_{2p}$; in practice it is dominated by the metastable $2s$ state.

2.1 Rate coefficients

The forward collisional rates are taken from Christie et al. Table 2 (Janev et al. 2003); the reverse (super-elastic) rates follow by detailed balance using the statistical weights $g_{1s} = 2$, $g_{2s} = 2$, $g_{2p} = 6$. With $t_4 \equiv T/10^4 \text{ K}$:

Process	Coefficient	Expression [$\text{cm}^3 \text{ s}^{-1}$]
Case-B recomb. (Draine 2011)	α_B	$2.54 \times 10^{-13} t_4^{-0.8163-0.0208 \ln t_4}$
$\rightarrow 2s$ (incl. cascade)	α_{2s}	$(0.282 + 0.047 t_4 - 0.006 t_4^2) \alpha_B$
$\rightarrow 2p$	α_{2p}	$\alpha_B - \alpha_{2s}$
$1s \rightarrow 2s$ excitation	$C_{1s \rightarrow 2s}$	$1.21 \times 10^{-8} t_4^{-0.455} e^{-118400/T}$
$1s \rightarrow 2p$ excitation	$C_{1s \rightarrow 2p}$	$1.71 \times 10^{-8} t_4^{-0.077} e^{-118400/T}$
$2s \leftrightarrow 2p$ ℓ -mix	$C_{2s \rightarrow 2p}$	$6.21 \times 10^{-5} (\ln(T/T_0) - \gamma) / \sqrt{T}$
$2s \rightarrow 1s$ (super-elastic)	$C_{2s \rightarrow 1s}$	$1.21 \times 10^{-8} t_4^{-0.455} (g_{1s}/g_{2s})$
$2p \rightarrow 1s$ (super-elastic)	$C_{2p \rightarrow 1s}$	$1.71 \times 10^{-8} t_4^{-0.077} (g_{1s}/g_{2p})$
$2p \rightarrow 2s$ ℓ -mix	$C_{2p \rightarrow 2s}$	$C_{2s \rightarrow 2p} (g_{2s}/g_{2p})$

with $T_0 = 1.02 \text{ K}$ and $\gamma = 0.57721$ (Euler–Mascheroni). The reverse rates are written in their analytically-cancelled form: the Boltzmann factor $e^{+118400/T}$ implied by detailed balance cancels the $e^{-118400/T}$ inside $C_{1s \rightarrow 2s}/C_{1s \rightarrow 2p}$, which avoids a numerical $0 \times \infty$ at the very low temperatures that can occur at the base of the post-processed profile.

2.2 Ly α pumping and the $J_{\text{Ly}\alpha}$ input

The $1s \rightarrow 2p$ pumping rate is $B_{1s \rightarrow 2p} J_{\text{Ly}\alpha}$, with the Einstein coefficient in the mean-intensity convention

$$B_{1s \rightarrow 2p} = \frac{g_{2p}}{g_{1s}} \frac{c^2}{2h\nu_{\text{Ly}\alpha}^3} A_{2p \rightarrow 1s} \simeq 8.55 \times 10^9 \text{ cgs},$$

and the stimulated-emission rate $B_{2p \rightarrow 1s} J_{\text{Ly}\alpha}$ with $B_{2p \rightarrow 1s} = (g_{1s}/g_{2p}) B_{1s \rightarrow 2p}$. Here $J_{\text{Ly}\alpha}$ is the **angle-averaged mean intensity J_ν at Ly α line center**, in cgs units $\text{erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1} \text{ sr}^{-1}$, so that $B_{1s \rightarrow 2p} J_{\text{Ly}\alpha}$ has units of s^{-1} .

$J_{\text{Ly}\alpha}(r)$ is obtained in one of two ways.

(1) From a file. If `Jlya_file` points to an existing two-column text file, $J_{\text{Ly}\alpha}(r)$ is read from it:

column 1	r/R_p	(same radial coordinate as the EXHALE output)
column 2	$J_{\text{Ly}\alpha}$	$[\text{erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1} \text{ sr}^{-1}]$

linearly interpolated onto the EXHALE grid and clamped to the endpoints. A template ships as `inputdata/Jlya.txt.example`.

(2) Simple estimate (default; Huang et al. 2017). If no file is given, $J_{\text{Ly}\alpha}$ is estimated following Huang, Arras, Christie & Li (2017, ApJ 851, 150), Eq. (6) and the surrounding text,

$$J_{\text{Ly}\alpha}(r) \simeq 0.1 \frac{F_{\text{LyC}}}{\Delta\nu_D(r)}, \quad \Delta\nu_D = \nu_{\text{Ly}\alpha} \frac{1}{c} \sqrt{\frac{2k_B T}{m_p}},$$

where $\Delta\nu_D$ is the local Ly α Doppler width and F_{LyC} is the *deposited* (absorbed) Lyman-continuum energy flux. The prefactor encodes the assumption that each LyC photoionization is balanced by a recombination producing one Ly α photon; Huang et al. identify F_{LyC} with the total LyC flux absorbed inside the atmosphere ($2.6 \times 10^4 \text{ erg cm}^{-2} \text{ s}^{-1}$ for HD 189733b, page 11).

F_{LyC} from the actual stellar input. Rather than a fixed number, EXHALE_transit.py computes the absorbed LyC flux from the EXHALE inputs:

$$F_{\text{LyC}} = \underbrace{\xi \frac{10^{L_{\text{EUV}}}}{4\pi a^2}}_{\text{incident stellar LyC}} \times \underbrace{(1 - e^{-\tau_{\text{LyC}}})}_{\text{fraction absorbed}}, \quad \tau_{\text{LyC}} = \sigma_{\text{LyC}} N_{\text{H I}},$$

where L_{EUV} (the EUV-band luminosity, $E > 13.6 \text{ eV}$) and the orbital distance a are read from `input.inp`, $N_{\text{H I}}$ is the vertical neutral-hydrogen column from the EXHALE profile, and $\sigma_{\text{LyC}} \simeq 6.3 \times 10^{-18} \text{ cm}^2$. The factor ξ is the day-night / 2D flux dilution, taken to match ATES's "2D approximate method" (`Rate/2` $\rightarrow$ 0.5, `Rate/4` $\rightarrow$ 0.25, else 1; read from `input.inp`, cf. the ξ factor of Christie+2013 / Huang+2017). For the optically-thick atomic layer $\tau_{\text{LyC}} \gg 1$, so the absorbed fraction $\rightarrow 1$ and $F_{\text{LyC}} \rightarrow \xi \times$ the incident stellar LyC flux (e.g. $\xi = 0.5$, $\simeq 5.2 \times 10^2 \text{ erg cm}^{-2} \text{ s}^{-1}$ for the HD 209458b input). A fixed value can be forced via `F_LyC_override` and the dilution via `xi_override`.

This estimate is a uniform-illumination approximation: it omits the decline of $J_{\text{Ly}\alpha}$ deep in the atmosphere captured by the full radiative-transfer treatment (Huang et al. Eq. 7) and the direct stellar Ly α component; for a self-consistent profile, supply `Jlya_file`.

2.3 $n = 2$ photoionization Γ_{2s}, Γ_{2p} (Balmer continuum)

The $n = 2$ levels are photoionized by the stellar Balmer continuum ($E > 3.4$ eV, $\lambda < 3647$ Å); this depletes n_2 and is included in Eqs. (12)–(13) through Γ_{2s}, Γ_{2p} . The Balmer-continuum field is outside EXHALE's XUV scope, so it is estimated from a diluted stellar blackbody: for a stellar effective temperature T_{star} and radius R_{star} ,

$$\Gamma_2 = \int_{\nu_2}^{\infty} \frac{F_{\nu}}{h\nu} \sigma_2(\nu) d\nu, \quad F_{\nu} = \pi B_{\nu}(T_{\star}) \left(\frac{R_{\star}}{a} \right)^2, \quad \sigma_2(\nu) = \sigma_{2,\text{th}} \left(\frac{\nu_2}{\nu} \right)^3,$$

with ν_2 the $n = 2$ threshold and $\sigma_{2,\text{th}} \simeq 1.4 \times 10^{-17} \text{ cm}^2$, and $\Gamma_{2s} \simeq \Gamma_{2p} \simeq \Gamma_2$. This reproduces Huang et al.'s Table 2 values ($\sim 20\text{--}26 \text{ s}^{-1}$) for a K dwarf, and is substantially larger for a hotter G dwarf (e.g. $\sim 140 \text{ s}^{-1}$ for HD 209458's $T_{\star} \simeq 6065 \text{ K}$), which suppresses n_2 and hence the Balmer-line depth. Setting $T_{\text{star}} \leq 0$ falls back to the manual constants $\text{Gamma}_{2s}/\text{Gamma}_{2p}$ (default 0).

3. H α and H β optical depth and transmission

Once $n_2(r)$ is known, the H α transmission is computed with the same LOS integration as Ly α . Along a chord at impact parameter b ,

$$\tau_{\nu}(b) = \int n_2(r) \sigma_{\text{H}\alpha}(\nu, r) d\ell, \quad \sigma_{\text{H}\alpha} = \frac{\sqrt{\pi} e^2}{4\pi\epsilon_0 m_e c} \frac{f_{23}}{\Delta\nu_D} H(a, x),$$

where $H(a, x) = \text{Re}[w(x + ia)]$ is the Voigt function (Faddeeva w), $\Delta\nu_D = \nu_{\text{H}\alpha} v_{\text{th}}/c$ uses the proton thermal speed, and the line-of-sight bulk velocity is folded into x exactly as for the existing lines. Atomic data: $\lambda_{\text{H}\alpha} = 6562.8$ Å (air), $f_{23} = 0.6407$, $A_{3 \rightarrow 2} = 4.41 \times 10^7 \text{ s}^{-1}$ (the line is strongly Doppler-dominated, so the damping wing is negligible). The disk-averaged transmission, instrument convolution and planet-rotation convolution reuse the existing code paths.

H β ($n = 2 \rightarrow n = 4$). The same $n = 2$ population also produces H β at $\lambda_{\text{H}\beta} = 4861.35$ Å ($f_{24} = 0.119$, $A_{4 \rightarrow 2} = 8.42 \times 10^6 \text{ s}^{-1}$). It is computed with an identical LOS loop and plotted/printed alongside H α . Because both lines are optically thick (saturated), the H α /H β depth ratio is much smaller than the optical-depth ratio (~ 7.3): e.g. on the converged HD 209458b run, $\text{H}\alpha \simeq 3.2\%$ and $\text{H}\beta \simeq 2.0\%$ with TPM's default star, and $\simeq 0.41\% / 0.26\%$ with HD 209458's actual $R_{\star}, T_{\star}$ (see the comparison notebook).

4. Implementation in EXHALE_transit.py

System parameters. The planet and the star are taken from the `input.inp` of the run directory, matched by label exactly as `input_read.f90` does. Besides R_p , M_p , T_{eq} , a , $M_{\star}$, L_{EUV} and the 2-D approximate method, this includes the stellar radius (Stellar radius [R_sun]) and effective temperature (Stellar Teff [K]), which set the transit normalization $A_{\star}$, the $R_{\star}/R_p$ cap on the absorbing annulus, and the diluted-blackbody Balmer continuum of §2.3. The planet spin period has no `input.inp` key: a close-in giant is tidally locked, so it defaults to the Keplerian orbital period $P_{\text{orb}} = 2\pi \sqrt{a^3/[G(M_{\star} + M_p)]}$. Each of the three is resolved as `EXHALE_TRANSIT_RSTAR_RSUN / _TSTAR / _ROTP` environment override `> ./input.inp >` built-in default, and the resolved values and their sources are printed at startup.

- `n2_populations(T, n1s, ne, Jlya, G2s, G2p)` solves the 2×2 system above and returns (n_{2s}, n_{2p}, n_2) in cm^{-3} .

The Ly α -pump constants $B_{1s \rightarrow 2p}$, $B_{2p \rightarrow 1s}$ are precomputed.

- $J_{\text{Ly}\alpha}$ is read from `Jlya_file` if present, otherwise built from the Huang et al. (2017) estimate (§2.2); n_2 is then built into the symmetric (inverted+normal) radial profile `data_n2` (in m^{-3}), exactly like `data_nHI`.
- A new wavelength grid `l_nde_Ha` and an H α LOS loop fill `exp_tau_Ha`; the disk average, instrument convolution (`Instr_res_Ha`) and rotation convolution produce the plotted curves.
- The H α branch always runs (`do_Ha=True`); with the default `Jlya_file=''` it uses the Huang et al. (2017) estimate for $J_{\text{Ly}\alpha}$, with F_{LyC} taken as the absorbed stellar LyC flux (§2.2).
- **Compatibility fix.** `Ion_species.txt` in EXHALE carries trace-metal columns (16 columns total), so the reader was changed to `usecols=range(7)` to take only r and the six H/He species. This also remains correct for the old 13-column (C/O-only) and 7-column (H/He-only) outputs.

4.1 Outputs

Every line carries one key — He10830, Lya, Halpha, Hbeta, MgII, CaII, NaI — and both products of a line are named from that key and land in the run directory:

Product	Name	Default
model curve	<path>/tpm_<line>.txt	always written
figure	<path>/<prefix><line>.png (+ .pdf)	off; set EXHALE_TRANSIT_FIG_PREFIX

A curve file has the columns `lambda[A]`, T_λ (theoretical), T_λ (instrument-convolved), T_λ (+rotation), so the excess absorption in percent is $(1 - T_\lambda) \times 100$. The set follows the run: a metals-off run writes no metal curves. EXHALE_TRANSIT_SAVE_PREFIX decorates the curve name exactly as the figure prefix decorates the figure name, and both are resolved against the run directory, so an absolute prefix writes elsewhere and nothing depends on the working directory.

5. Validation

Run end-to-end on the converged HD 209458b EXHALE output (`output/*_adv.txt`), the module is stable and NaN-free and yields clean H α and H β absorption lines at 6562.8 and 4861.4 Å. With the Huang $J_{\text{Ly}\alpha}$ estimate ($\xi = 0.5$, $F_{\text{LyC}} \simeq 5.2 \times 10^2$, Γ_2 from $T_* = 6065 \text{ K}$), the depths are physically reasonable and the H α /H β ratio reflects saturation (both optically thick). The Ly α branch in the same run saturates ($\sim 100\%$ core absorption) and He I behaves as before. A worked comparison against the Jensen et al. (2012) HD 209458b observation is provided in the notebook `Halpna_compare_HD209458b.ipynb`.

6. Metal resonance doublets (Mg II h&k, Ca II H&K, Na I D)

The metal resonance lines are the key NUV/optical diagnostics of metal-enriched escaping atmospheres (Huang et al. 2023). They absorb out of the *ion ground state*, whose excited fine-structure levels are Boltzmann-negligible at $\sim 10^4 \text{ K}$, so the lower-level density is the ion density itself ($n_{\text{lower}} \simeq n_{\text{ion}}$) — no non-LTE population calculation is

needed. The ion-density profiles are read from the metal block of `Ion_species_adv.txt`; a metals-off run (all-zero columns) skips this section automatically.

6.1 Atomic data and implementation

Each species is treated as a **doublet**, with both components summed in a single wavelength window (Table 1; NIST f -values and A -coefficients). The optical depth uses exactly the H/He pipeline: spherical chords $\sqrt{r^2 - b^2}$ on the impact-parameter grid, LOS velocity $v_{\text{los}} = vx/r$, Voigt profiles via the Faddeeva function (vectorized over wavelength), trapezoidal τ , projected-area disk average, then the instrument convolution and the planet-rotation convolution with the depth-dependent effective radius $R_{\text{eff}} = R_p \sqrt{(h + \delta)/\delta}$ (capped at the impact-parameter boundary), $\delta = (R_p/R_\star)^2$. The three doublets are written under the same `tpm_<line>` naming as the H/He lines (§4.1), with the keys MgII, CaII, NaI.

Table 1: Metal resonance doublets in EXHALE_transit.py (NIST).

species	component	λ_0 [Å]	f	A_{21} [s $^{-1}$]	window / R_{instr}
Mg II	k	2796.352	0.608	2.60×10^8	2790–2810 Å / 3×10^4
	h	2803.531	0.303	2.57×10^8	
Ca II	K	3933.663	0.6267	1.47×10^8	3927–3975 Å / Instr_res_Ha
	H	3968.469	0.3116	1.40×10^8	
Na I	D2	5889.951	0.641	6.16×10^7	5884–5902 Å / Instr_res_Ha
	D1	5895.924	0.320	6.14×10^7	

Alongside the spectra, the script prints the validated *single-component* depth table (line-center and 4 Å-band percentages for Mg II 2796, Ca II K, Na I D2), which is the quantity used for the Huang et al. (2023) comparisons; the doublet peak differs from the single-component value only by the partner-line wing (e.g. 3.75 vs. 3.72% for Mg II on WASP-121b). Two conventions matter when comparing with Huang: their table lists the effective transit radius $R_{\text{eff}}/R_\star = \sqrt{(R_p/R_\star)^2 + h}$, not a percent absorption, and Mg II is quoted in a 4 Å NUV bin while the optical lines are line-center (line-center vs. 4 Å differ by $\sim 3\times$ for the optically thick Mg II core).

For a Roche-lobe run the 3-D equipotential reconstruction (`geometry = 'triaxial'`, `roche_recon.py`) replaces the spherical chords by the full Roche geometry with an LOS velocity for each sector $v_{\text{sub}}(r_{\text{eff}})x/r - \Omega y$ (wind + tidally locked rotation) and reports the same depth table.

6.2 Example spectra

Figure 1 shows the three doublet spectra for the WASP-121b worked example (`backup/regression/tpm_wasp/`, Roche Newton solution, metals on). The optically thick cores saturate, so the two doublet members reach similar depths despite the 2:1 oscillator-strength ratio; the rotation convolution mainly shallows and broadens the narrow Na I D cores.

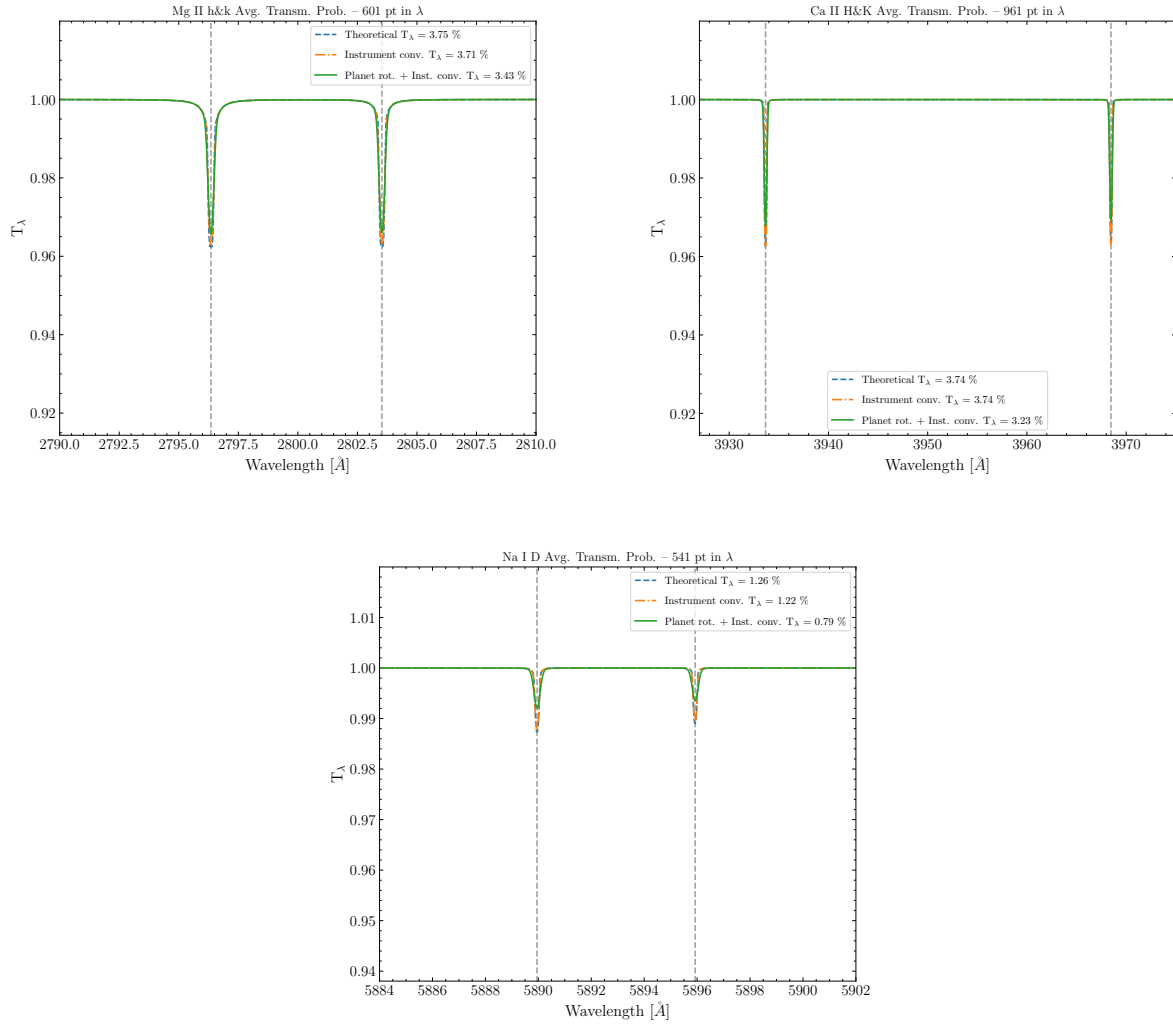


Figure 1: Metal resonance doublet transmission spectra for WASP-121b (backup/regression/tpm_wasp/): Mg II h&k (top left), Ca II H&K (top right), and Na I D (bottom). Each panel shows the theoretical, instrument-convolved, and instrument+rotation-convolved transmission, as for the H/He lines.

7. Caveats

1. **$n = 2$ photoionization.** Γ_{2s}, Γ_{2p} are now estimated from a diluted stellar-blackbody Balmer continuum (§2.3, parameter T_star); this is a single blackbody approximation to the true stellar near-UV spectrum. Setting T_star ≤ 0 reverts to the manual constants (default 0).
2. **Electron density.** $n_e = n_{\text{HII}} + n_{\text{HeII}} + 2n_{\text{HeIII}}$ from EXHALE is used (Christie assume $n_e = n_p$); the difference is small.
3. **$J_{\text{Ly}\alpha}$ convention.** The input must be the mean intensity J_ν at line center in cgs (§2.2). A different convention (energy density, frequency-integrated, etc.) requires rescaling the pump constant.
4. **Wavelength/damping.** $\lambda_{\text{H}\alpha}$ is the air value 6562.8 Å; the natural width is negligible relative to the thermal Doppler width, so the line is essentially Gaussian.

Reference

- Christie, D., Arras, P., & Li, Z.-Y. 2013, ApJ, 772, 144 — “H α Absorption in Transiting Exoplanet Atmospheres” (references/Christie_2013ApJ_772_144.pdf).
- Huang, C., Arras, P., Christie, D., & Li, Z.-Y. 2017, ApJ, 851, 150 — “A Model of the H α and Na Transmission Spectrum of HD 189733b” (references/Huang_2017_ApJ_851_150.pdf); the $J_{\text{Ly}\alpha}$ estimate is their Eq. (6) and related text.
- Jensen, A. G., et al. 2012, ApJ, 751, 86 — transit H α spectroscopy of HD 209458b and HD 189733b (comparison target for the HD 209458b notebook).